# AI-compute supply chain

Chart pack based on analysis of data from UN Comtrade + TDM + Trade Atlas, from which we construct a bilateral monthly panel of HS6 trade flows capturing the global AI-compute supply chain (60 HS6 codes, 2017-01 to 2026-04).

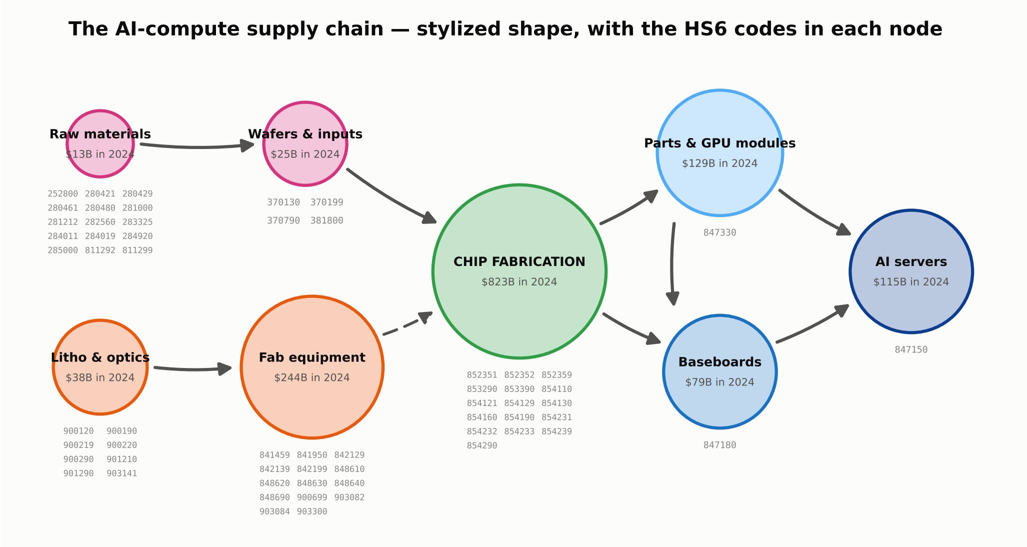
2026-08-20

# 1. The supply chain in 2024

2024 annual trade flows from the Trade Atlas. We record all relevant HS6 codes, so its coverage extends beyond solely AI compute. China and Hong Kong are counted as one trade bloc (CHK) throughout.

# Topography of the supply chain

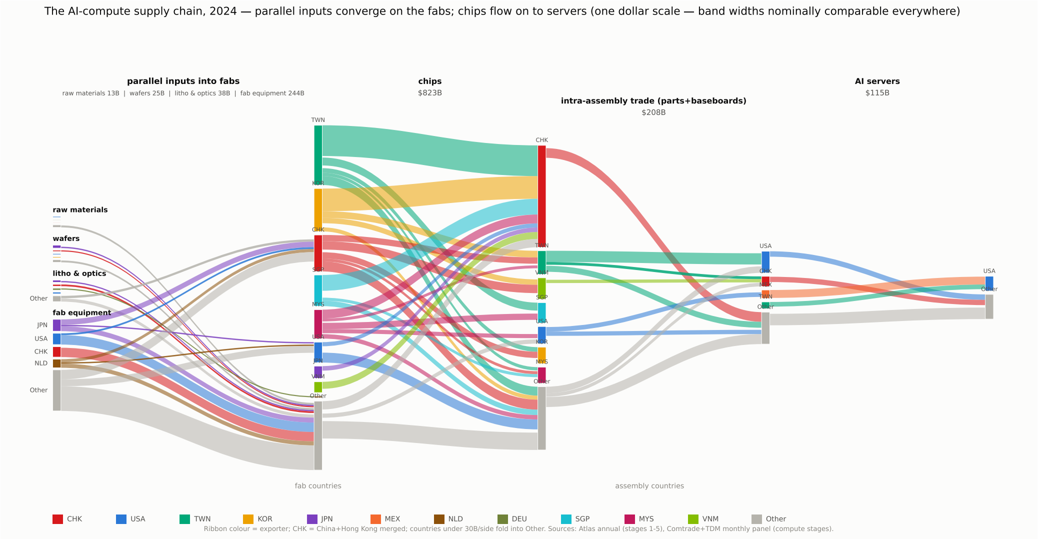
Two upstream inputs feed chip fabrication: capital equipment in orange is used to build manufacturing facilities, while raw materials and wafers in pink are consumed per chip. Green represents the actual chip fabrication, which feeds downstream assembly toward AI servers in blue. Circle size in proportion with the log of 2024 USD total trade value.




Two input branches, one output line. Materials (pink) -- silicon, chemicals, wafers -- are used up with every chip made (solid arrows). Equipment (orange) -- lithography and other fab tools -- is investment in capacity: in Taiwan and Korea, equipment imports rise six to twelve months before chip exports (dashed arrow). Korean memory and Taiwanese chips are packaged together, usually in Taiwan. Chip design earns as a service and crosses no border. Circle diameters grow with the log of 2024 world trade. Blue marks the three codes of our monthly panel (the Fed AI-compute basket). A few niche inputs (lasers, vacuum pumps and valves, chip substrates) fall outside the 60 codes. Codes: OECD semiconductor mapping + Fed basket (docs/data.md, section 1).

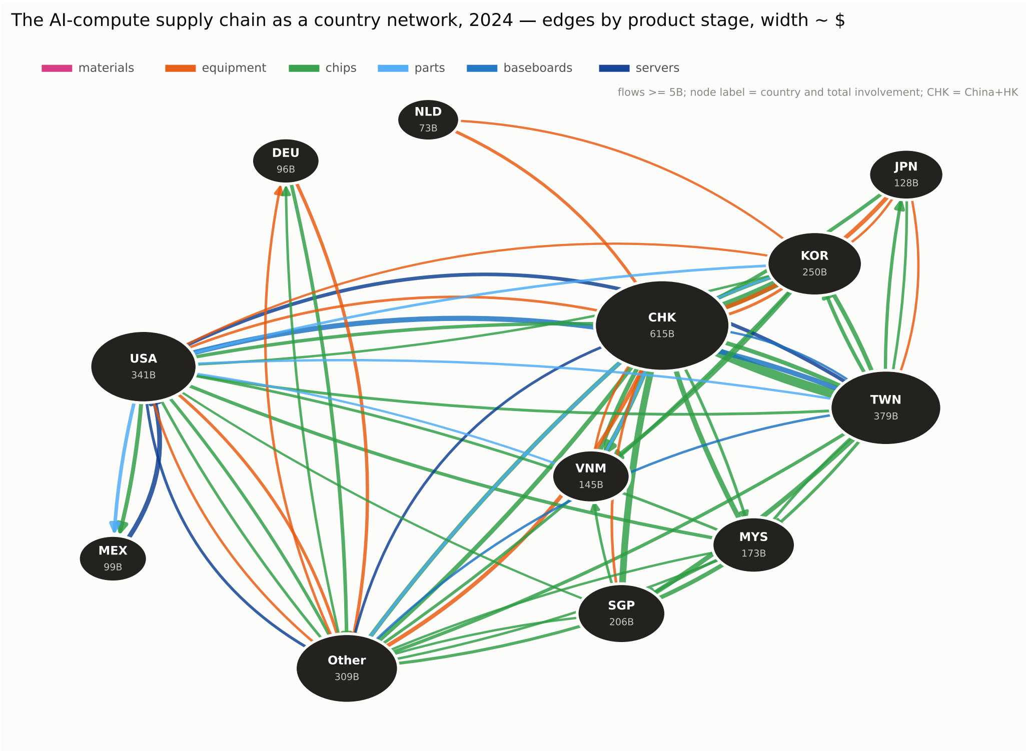
# The chain in bilateral trade flows

Width of band = 2024 USD trade. Small flows are grouped into other (grey).



# The chain as a graph network

Countries are represented as nodes and directional flows as edges, colored by stage.



# 2. The chain in stages

2024 annual trade flows from the Trade Atlas. We estimate a matrix factor model to summarize trade flows by exporting and importing hubs.

## The matrix factor model

Per stage, the bilateral export matrix, $X$, has $i$-$j$th entry giving the export value from country $i$ to country $j$ in 2024. The Matrix Factor Model (MFM) for $X$ is given by

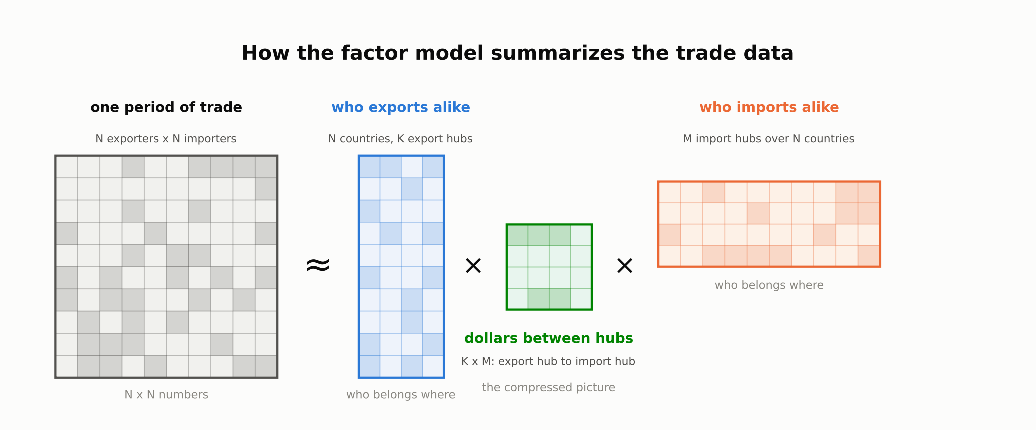
$$X = R\,F\,C' + e,$$

for $N$ countries, $K$ export hubs indexed by $k$, and $M$ import hubs indexed by $m$:

- $R$ $(N \times K)$ are the **export loadings**: the weight of each country in each export hub.
- $C$ $(N \times M)$ are the **import loadings**: the weight of each country in each import hub.
- $F$ $(K \times M)$ are the **matrix factors**: $F_{km}$ is the flow from hub $k$ to hub $m$.
- $e$ $(N \times N)$ are the idiosyncratic errors.

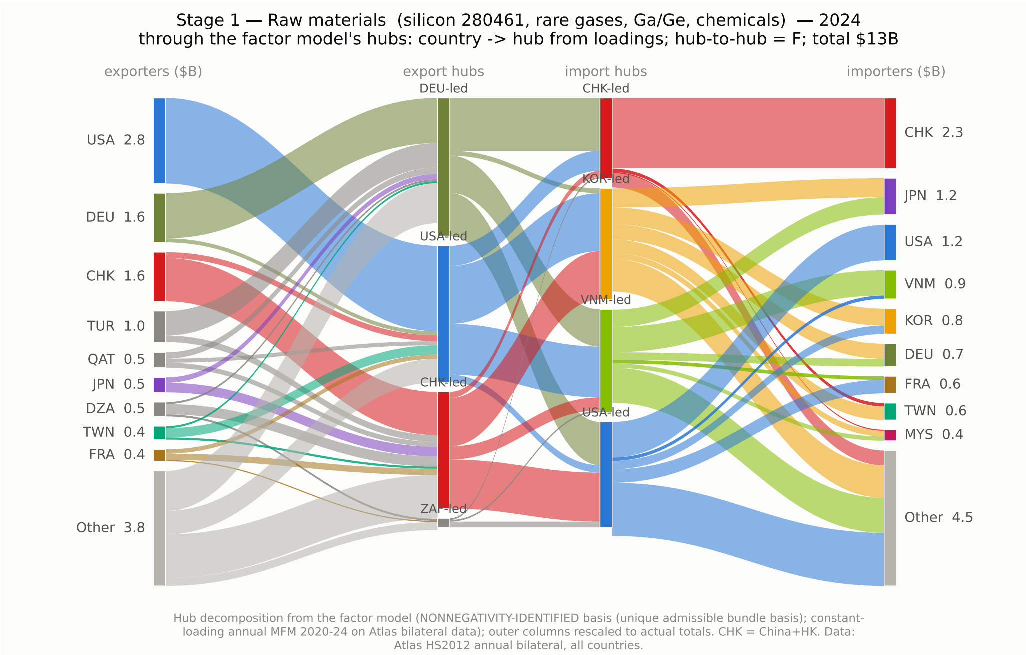
# The same model as a picture

Similar to a PCA model, which summarizes co-movement in a vector as loadings times a low-dimensional vector of factors, the MFM does the same for a matrix: it summarizes $X$ by a low-dimensional matrix of factors with two sets of loadings.



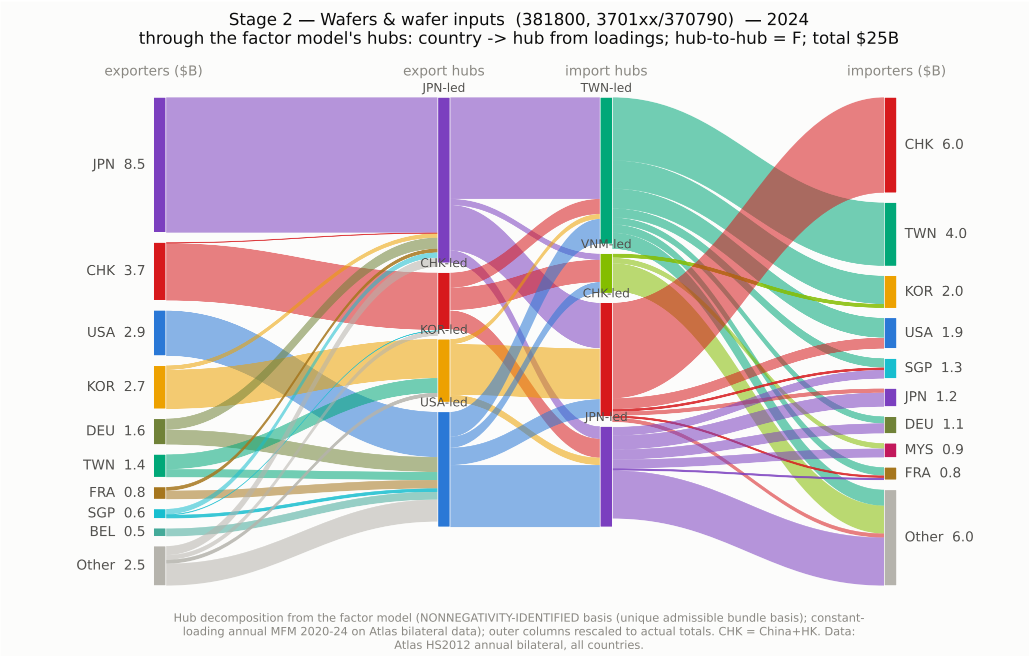
# Stage 1 – raw materials

Left to right: exporters to export hubs to import hubs to importers. Raw materials trade is relatively small (13B) and diffused across countries.



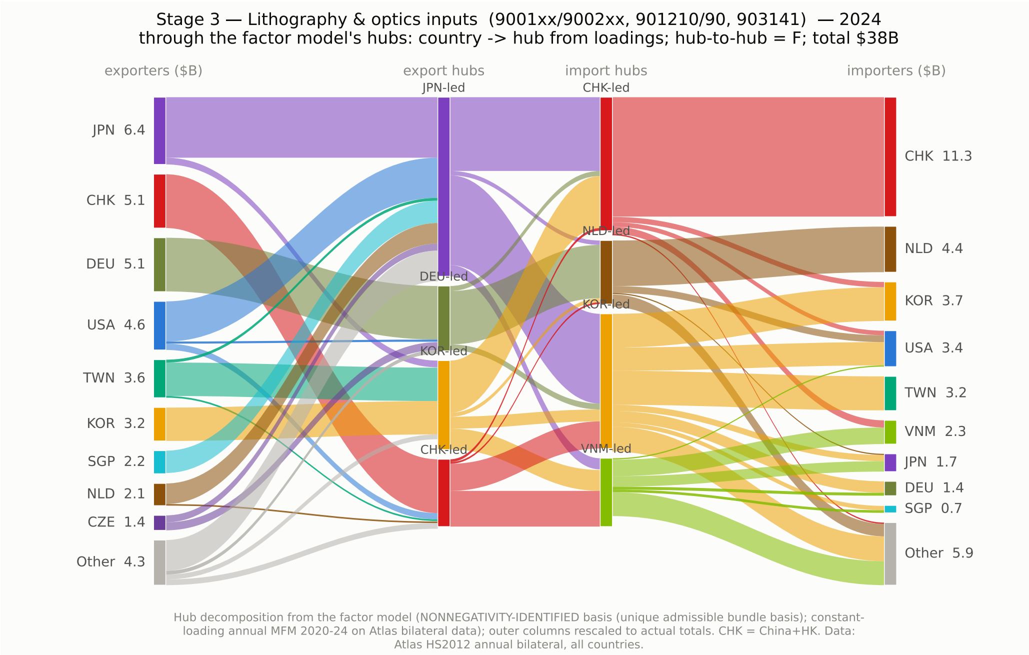
# Stage 2 – wafers and wafer inputs

Japan is the largest exporter, about a third of the 25B world total.



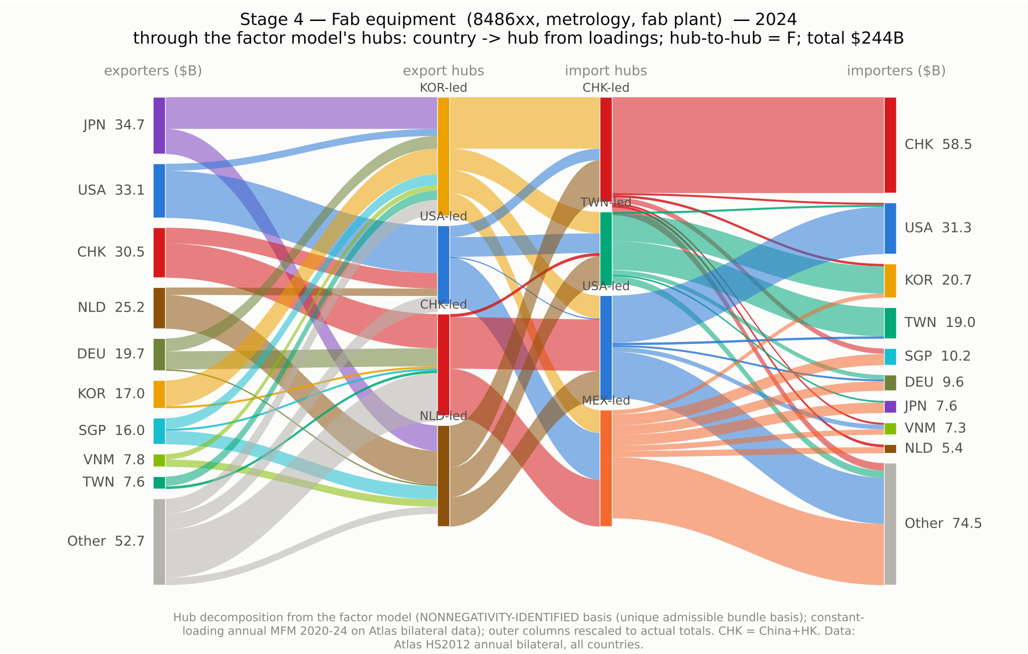
# Stage 3 – lithography and optics inputs

Lenses, optical elements, electron microscopes and wafer inspection tools. No dominant supplier: Japan, the China bloc, Germany and the United States each export 5 to 6B.



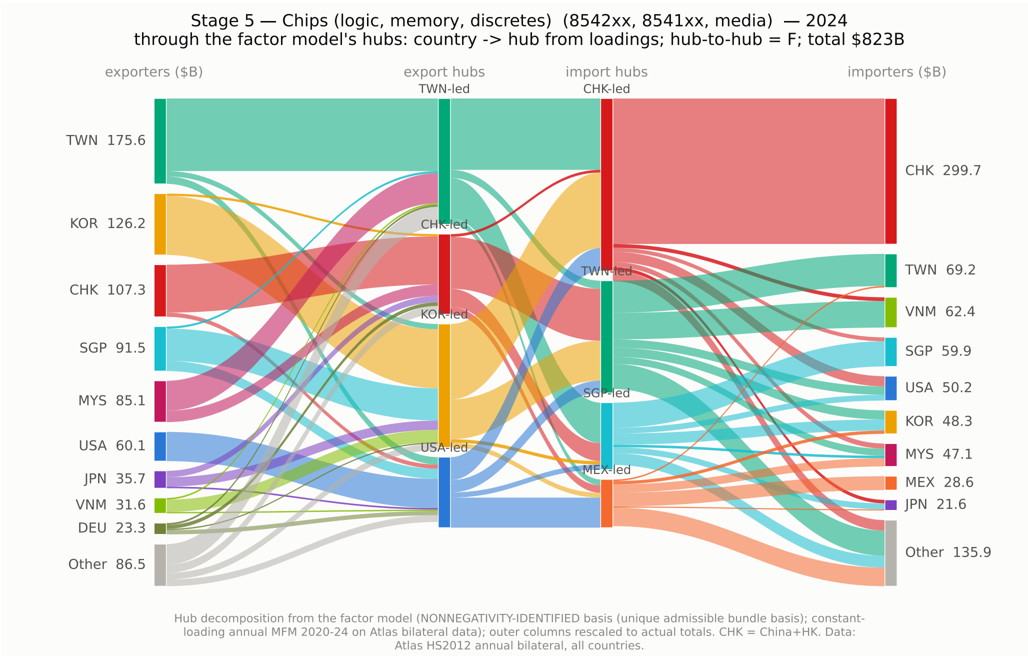
# Stage 4 – fab equipment

Machines, not materials: countries buy these to build capacity, and their imports rise 6-12 months before chip exports.



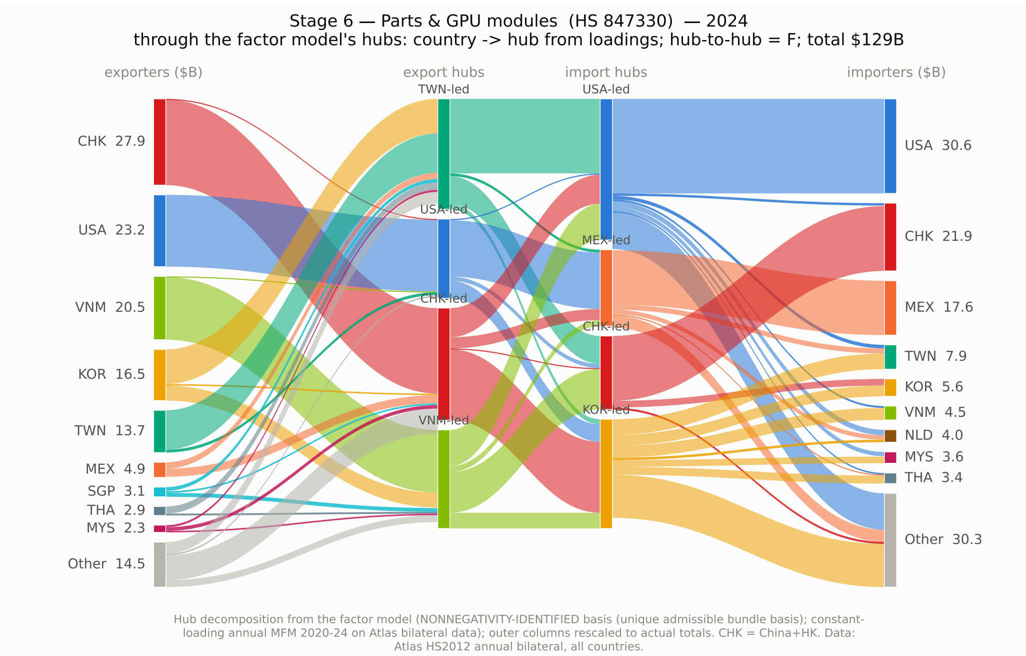
# Stage 5 – chips

The biggest stage, and the whole electronics industry – phones, cars and PCs, not just AI. Most chips flow to Asian assembly; the China bloc alone absorbs 36 percent of world chip imports.



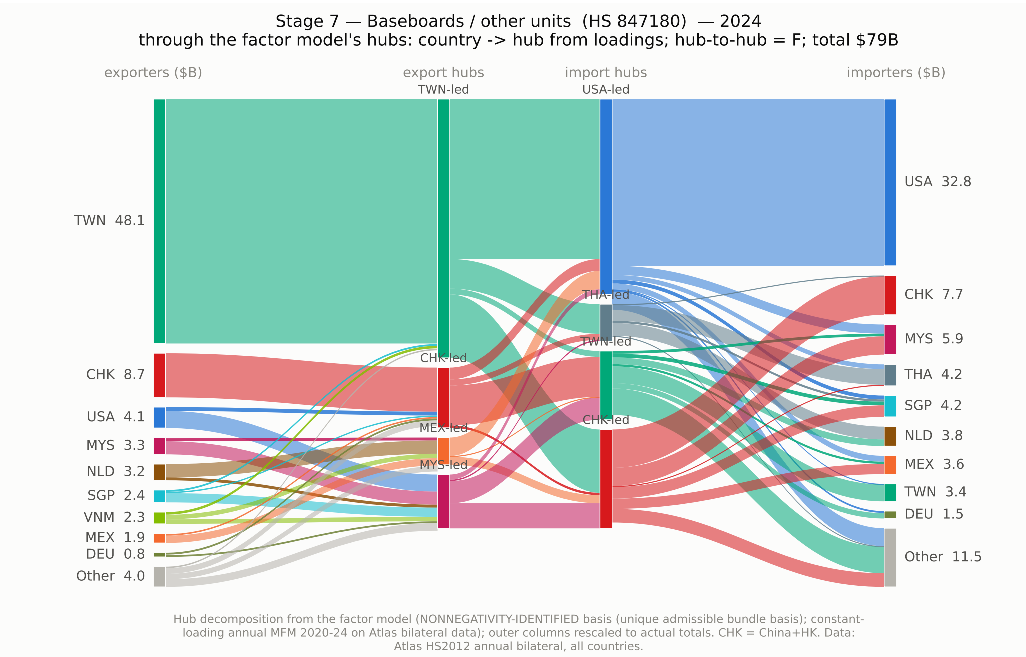
# Stage 6 – parts and GPU modules (847330)

The China bloc exports the most by value (28B in 2024), ahead of the United States and Vietnam; the United States is the largest buyer (31B), then the China bloc and Mexico.



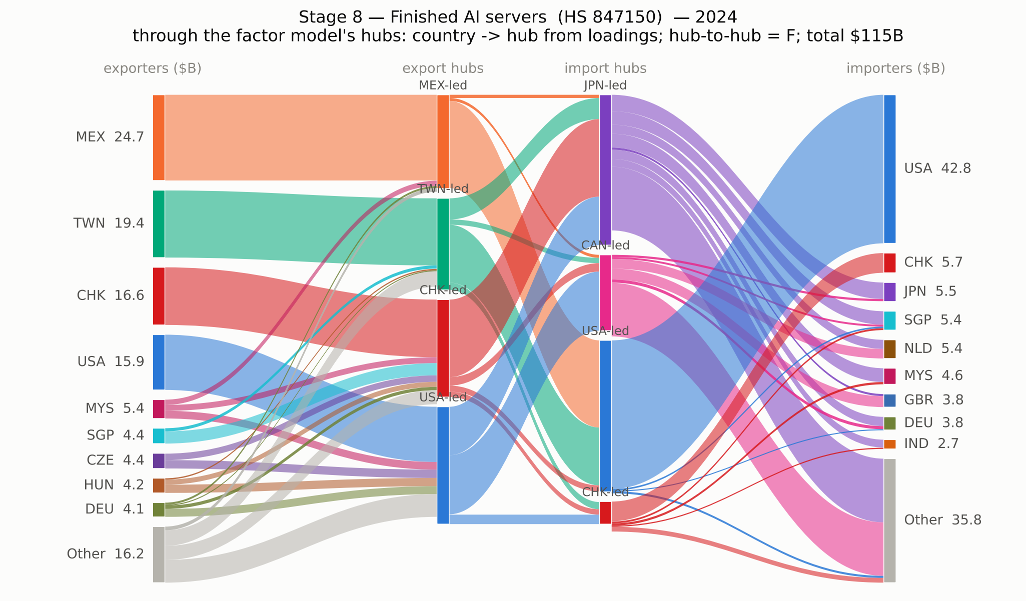
# Stage 7 – baseboards (847180)

The most concentrated stage: Taiwan exports 48B of the 79B total in 2024, five times the next exporter, and the United States buys 42 percent of world imports.



# Stage 8 – AI servers (847150)

Mexico (25B) and Taiwan (19B) assemble and ship; the United States buys 43B, 37 percent of world imports in this code.



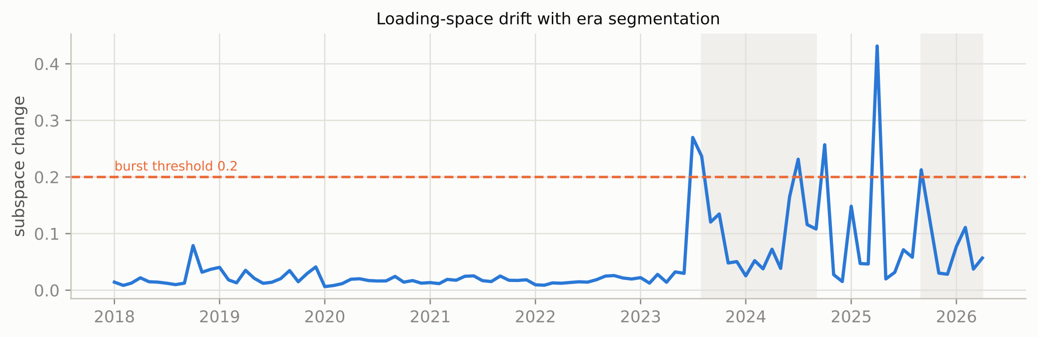

Hub decomposition from the factor model (NONNEGATIVITY-IDENTIFIED basis (unique admissible bundle basis); constant-loading annual MFM 2020-24 on Atlas bilateral data); outer columns rescaled to actual totals. CHK = China+HK. Data: Atlas HS2012 annual bilateral, all countries.

# 3. Changes over time – AI-compute only

This section looks at the three downstream AI-compute HS6 codes jointly (parts/GPU modules, baseboards, AI servers), and estimates a time-varying MFM on monthly trade from 2017.

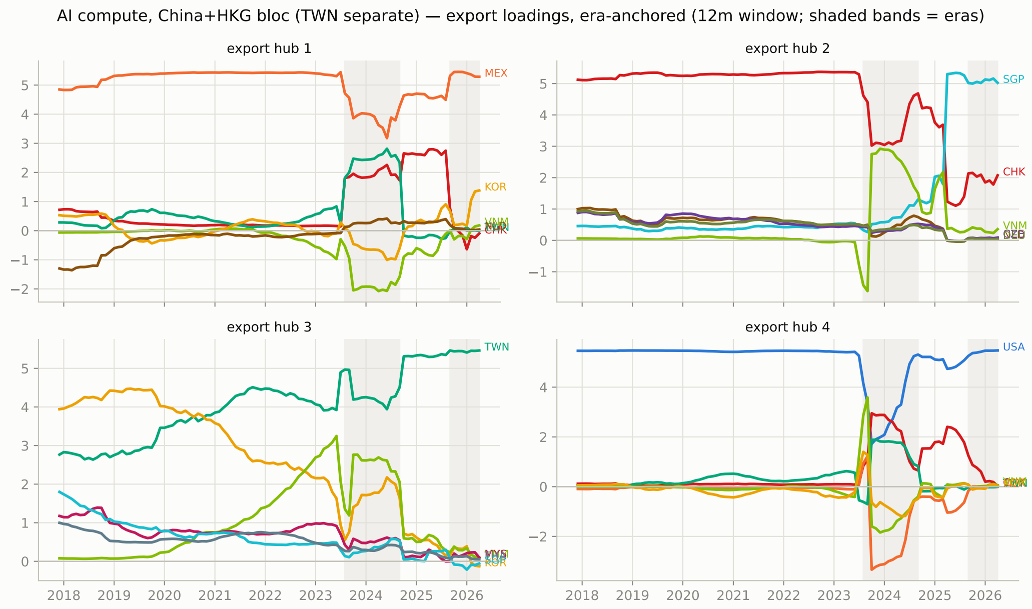
# When the network changed shape

The line plots the month-to-month change in time-varying export loadings. We see stability until Aug 2023 (AI demand rampup), Oct 2024 (memory export controls signaled), and Sep 2025 (Nvidia GB300 production).



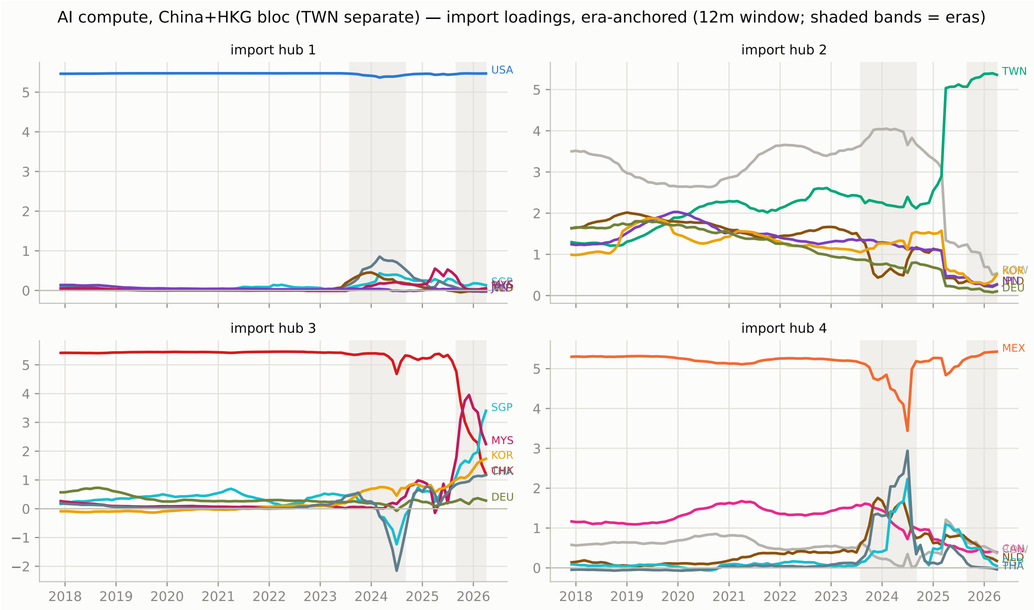
# Who is in each export hub

For export hub 2, Singapore swaps for China's export share in October 2024, perhaps reflecting transshipment to circumvent tariffs or export controls.



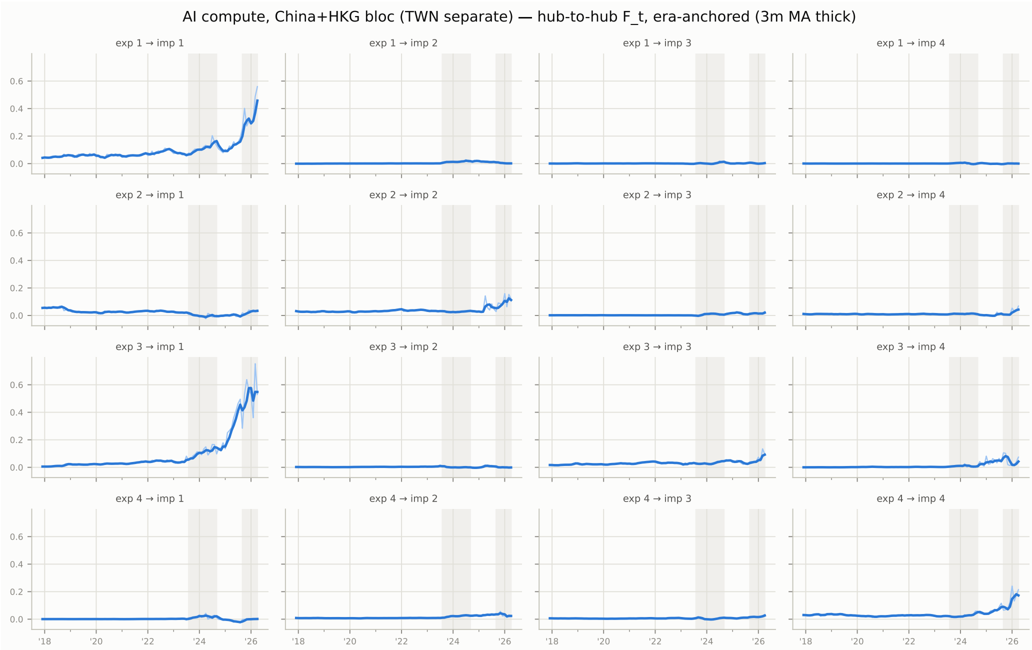
# Who is in each import hub

For import hub 2, Taiwan comes to dominate from late 2024, while import hub 3, previously China-led, becomes diffuse from late 2025.



# Trade between hubs, month by month

Three of the sixteen channels carry the AI boom: Taiwan to the US (export hub 3 to import hub 1), and both legs of the US-Mexico assembly loop (export hub 1 to import hub 1, export hub 4 to import hub 4).

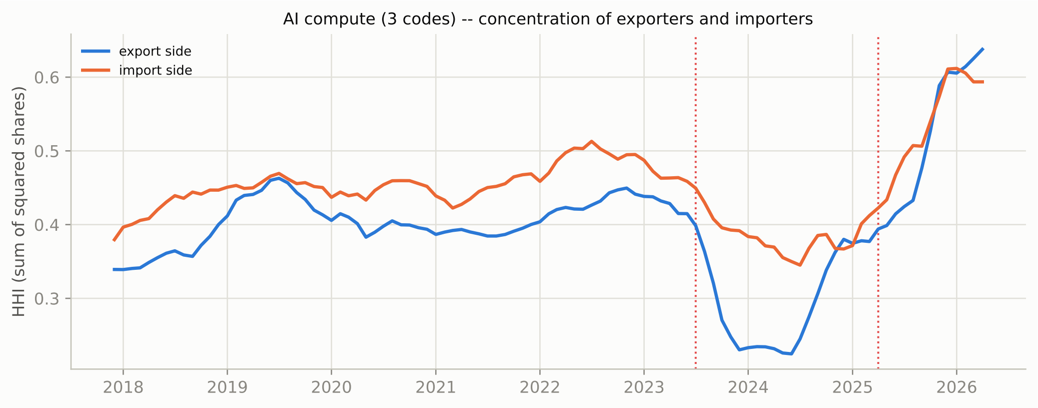


# 4. Concentration & fragmentation – AI-compute only

Monthly proxies for concentration and fragmentation calculated on the monthly time-varying MFM.

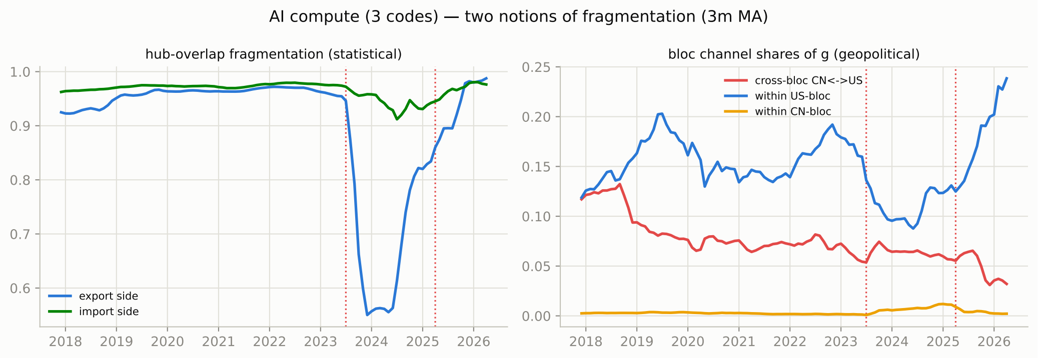
# Export and import concentration – AI-compute

Concentration is proxied by HHI: for exporters/importers, the HHI is computed within each hub and then averaged across hubs, weighted by trade.



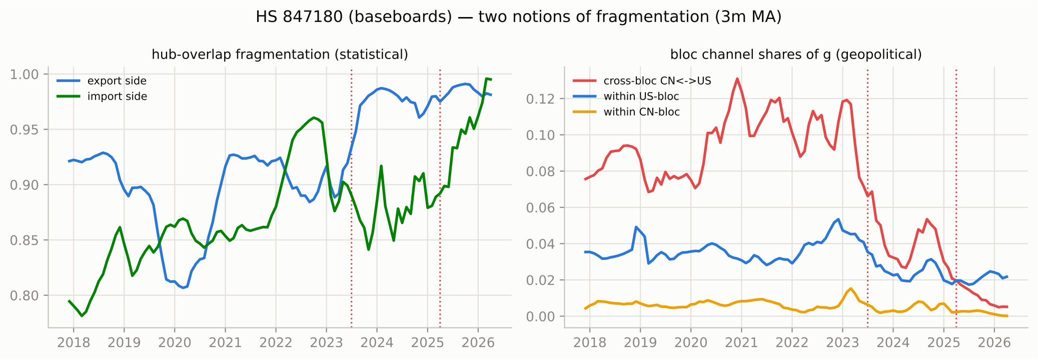
# Two proxies for fragmentation – AI-compute

Left: do the export and import hubs overlap with one another (based on the tvMFM estimation); right: the share of trade within and across the China-HK vs the US (US, Mexico, Canada) blocs.



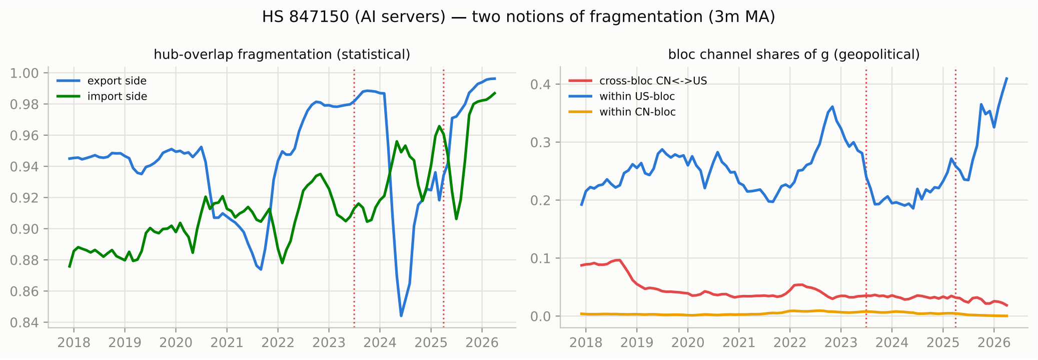
# Baseboards: the clearest fragmentation

The cross-bloc share collapses about 20x from its 2021 peak – this is the HS6 code most exposed to export controls.



# Servers: the US bloc picks up

Cross-bloc share falls from 0.09 to 0.02 while trade within the US bloc doubles to 0.40 – server assembly is consolidating inside one bloc.



# Parts: still diffused

The cross-bloc share halves but stays the highest of the three codes.